



- Extreme Climate Adaptability Full-load operation, 45°C without derating
- High Battery Compatibility Max. 415Acharge/discharge, supports three battery inputs
- Wide Grid Adaptability 154V~286V per phase input, 50/60Hz autoadaptation, built-in SPD
- High Surge&Unbalanced Output 2xload surge for 10s, 150% three-phase unbalanced output
- Flexible System Expansion Up to 8 units parallel, supports AC coupling for retrofit
- SiC + Full Aluminum +IP66High efficiency, lightweight, IP66 protection



# Three Phase Hybrid Inverter

TLH-HYB Series

| Model                                 | HYB-20KTLH-01-TVN                                                                                         | HYB-25KTLH-01-TVN |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------|-------------------|
| Inverter Input/Output(grid)           |                                                                                                           |                   |
| Rated Power(kw)                       | 20/20                                                                                                     | 25/25             |
| Rated Voltage                         | 380/400Vac, 3W+N+PE                                                                                       |                   |
| Rated Frequency(Hz)                   | 50/60                                                                                                     |                   |
| Rated Factor Range                    | 0.8 leading-0.8 lagging                                                                                   |                   |
| Switch Time(ms)                       | <20                                                                                                       |                   |
| Current(Maximum/Continuons)(A a.c.)   | 58/29                                                                                                     | 60/35             |
| Max Output Overcurrent Protection     | 400Vac/100Aac                                                                                             |                   |
| Battery                               |                                                                                                           |                   |
| Battery Type                          | Lithium/Lead-acid                                                                                         |                   |
| Battery Voltage Range(V d.c.)         | 40-60                                                                                                     |                   |
| Rated Battery Voltage(V d.c.)         | 48                                                                                                        |                   |
| Max Charge Current (V d.c.)           | 390                                                                                                       | 415               |
| Max Discharge Current (V d.c.)        | 390                                                                                                       | 415               |
| PV Input                              |                                                                                                           |                   |
| No.of MPPT Tracker/Strings            | 2/2+2                                                                                                     |                   |
| Max.PV Array Power(kW)                | 40                                                                                                        | 48                |
| Max DC Voltage(V d.c.)                | 1000                                                                                                      |                   |
| MPPT Voltage Range (V d.c.)           | 150-950Vdc                                                                                                |                   |
| Start-up Voltage(V d.c.)              | 180                                                                                                       |                   |
| Max.Input Current Ad.c.)              | 4*20                                                                                                      |                   |
| Isc PV(Absolute Maximum)(A d.c.)      | 4*30                                                                                                      |                   |
| AC Output(EPS)                        |                                                                                                           |                   |
| EPS Rated Output Voltage              | 380/400Vac, 3W+N+PE                                                                                       |                   |
| EPS Rated Output Frequency(Hz)        | 50/60                                                                                                     |                   |
| EPS Rated Output Active Power(kW)     | 20                                                                                                        | 25                |
| EPS Rated Output Apparent Power (kVA) | 20                                                                                                        | 25                |
| EPS Rated Output Current(Aa.c.)       | 29                                                                                                        | 35                |
| Overload Capacity (Off Grid)          | 100%600s/120%300s/150%60s/200%10s                                                                         |                   |
| Efficiency                            |                                                                                                           |                   |
| MPPT Efficiency                       | 99.90%                                                                                                    |                   |
| Euro-Efficiency                       | 97.20%                                                                                                    |                   |
| Max.Efficiency                        | 98.00%                                                                                                    |                   |
| Protection &Feature                   |                                                                                                           |                   |
| Insulation Monitoring                 | Yes                                                                                                       |                   |
| Residual Current Monitoring           | Yes                                                                                                       |                   |
| Parallel Function                     | Yes                                                                                                       |                   |
| Protection Degree                     | IP66                                                                                                      |                   |
| Certifications                        | IEC62109-12,IEC62477-1,IEC 61000-6-1,IEC 61000-6-3,EN 50549-1&-10,NRS 097-2-1,IEC61727,JIEC61683,IEC62116 |                   |
| Other Protection                      | Output over voltage,Output over current,Short circuit,Surge,Temperature,Anti-Islandmg                     |                   |
| General Parameterr                    |                                                                                                           |                   |
| Storage Temperature                   | -25℃~+60℃                                                                                                 |                   |
| Operating Temperature                 | -25℃~+60℃(Linely derating to 60%when exceed+45℃~+60℃)                                                     |                   |
| Humidity                              | 0-95%(Non-condensing)                                                                                     |                   |
| Altitude(m)                           | 4000(>2000 Derating)                                                                                      |                   |
| Noise(dB)                             | <75                                                                                                       |                   |
| Machine Dimensien(W*H*D)(mm)          | 448*660*265                                                                                               |                   |
| Machine Weight/N.W. (kg)              | 48                                                                                                        |                   |
| Display&Communication                 |                                                                                                           |                   |
| Display                               | LCD+APP                                                                                                   |                   |
| Interface                             | RS485/WIFI/CAN/DRM (Optional:Bluetooth/LAN/Meter)                                                         |                   |